James Davis Maxwell

Curriculum Vitae

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Education

University of Virginia, Charlottesville, VA

Ph.D. in Physics, December 2011

- Thesis Topic: *Measurement of g_2^p at Four-momentum Transfer of 2-6 GeV^2*
- Advisor: Donal B. Day
- Presidential Fellowship, 2004-2008

M.A. in Physics, May 2010

B.S. in Physics and Mathematics, May 2004

- *High Distinction*, Distinguished Majors Program
- Independent Study: *Wavelet Transforms for Denoising NMR Signals*

Positions

Thomas Jefferson National Accelerator Facility, Newport News, VA

Staff Scientist, Target Group, August 2015 to present

- Cryogenic & Polarized Target Development under Chris Keith
- Promoted to Staff Scientist III, April 2024

Massachusetts Institute of Technology, Cambridge, MA

Senior Postdoctoral Associate, Laboratory for Nuclear Science
June 2012 to August 2015

- Polarized ^{3}He Ion Source Development under Richard Milner
- SNS nEDM Cryogenics and DAQ under Robert Redwine

Postdoctoral Fellow, Harvard University Department of Physics
October 2012 to August 2015

- Visiting fellow, Dilution Refrigerator Cryogenics under Isaac Silvera

University of New Hampshire, Durham, NH

Postdoctoral Research Associate, Nuclear and Particle Physics Group
August 2011 to June 2012

- Solid Polarized Targets under Karl Slifer

Research Experience

Cryogenic and Polarized Target Development at Jefferson Lab

- Q-meter Prototyping, NMR System Design and Fabrication, Cryogenic Pulse Tube Test Stand Design and Fabrication, CLAS12 Polarized Target Development and Operation, Target Expert for PRad Experiment Internal Gas Target, Hall B Slow Controls Software Design and Implementation, Instruction of Postdocs, Graduate and Undergraduate Students
- Experimental Physics & Industrial Control System (EPICS) slow controls instrumentation software for Hall B Cryotarget, written in Python

A Program of Spin-dependent Electron Scattering using a Polarized 3He Target in CLAS12, JLab PR12-20-002

- Spokesperson, Target Conception and Leadership of Development, Construction of Helium-3 Polarizer

Search for Exotic Gluonic States in the Nucleus, JLab LOI12-16-006

- Spokesperson, Kinematics Investigation, Target Development

Polarized 3-Helium Development for RHIC, Brookhaven National Laboratory

- Project Leader, Test Design, Data Acquisition, Drafting, Magnetic Field Simulation, Laser Polarimeter Design and Fabrication, Wrote DOE Grant Quarterly Reports and Renewals, Instruction of Graduate and Undergraduate Students

SNS nEDM experiment, Oak Ridge National Laboratory

- Cryogenic Dilution Refrigerator Operation for Super-fluid Film Burner Tests, Data Acquisition Design, Ultra-Cold Neutron Storage Shifts at LANL

g2p and GEp experiments, JLab E08-027 and E08-007, Hall A, 2012

- Run Coordinator, Target Operator Trainer, Target Expert, Target Group Liaison

Spin Asymmetries on the Nucleon Experiment, JLab E07-003, Hall C, 2009

- Target DAQ, Target Expert, Target Operator Trainer, Analysis

Synergistic Activities

Thomas Jefferson National Accelerator Facility

- Experimental Readiness Review Committee Member: E12-06-113, E12-17-006, E12-17-006A, E12-06-106, E12-06-117, E12-09-016, E12-22-005
- Safety Response Excellence Recognition, August 2024

International Workshop on Polarized Sources, Targets and Polarimetry

- Local Organizer, September 2024

DOE Basic Needs Research Workshop on Laser Technology

- Panelist, July 2023

DOE Science Communications Summit

- Invited Presenter/Panelist, June 2023

Sony Pictures Entertainment

- Technical consulting for *Ghostbusters* (2016), including prop design and construction for laboratory sets, on-set consultation during filming, scientific background writing, and script consultation. Gave numerous interviews to news media, including an appearance on NPR's *Science Friday*. Received film credit.

Northeastern University Research, Innovation and Scholarship Expo (RISE)

- Judge, 2016-2020

Massachusetts Institute of Technology

- President's Committee on Radiation Protection, Postdoctoral Member, 2013-2015

PATENTS

"Digital Q-Meter for continuous-wave NMR," **J. Maxwell**, H. Dong, C. Keith, C. Cuevas. US Patent Number 11726153, (2023).

PUBLICATIONS

"EPICS for Small-Scale Laboratories with Python Soft IOCs," **J. Maxwell**, Journal of Instrumentation **21** (2026). doi.org/10.1088/1748-0221/21/03/P03030

"A New cw-NMR Q-meter for Dynamically Polarized Targets for Particle Physics," **J. Maxwell**, *et al.* Nuclear Instruments and Methods in Physics Research A **1087** (2026). doi.org/10.1016/j.nima.2026.171417

"Realizing the Scientific Program with Polarized Ion Beams at EIC," EPIOS Scientific Consortium, Accepted. arxiv.org/abs/2510.10794

"Polarizing 3He via metastability exchange optical pumping using a 1.2 mbar sealed cell at magnetic fields up to 5 T," P. Pandey, H. Lu, **J. Maxwell**, *et al.*, Nuclear Instruments and Methods in Physics Research A **1081** (2025). doi.org/10.1016/j.nima.2025.170870

"Horizontal 1 K refrigerator with novel loading mechanism for polarized solid targets," J. Brock *et al.*, Cryogenics **150** (2025). doi.org/10.1016/j.cryogenics.2025.104146

"Metastability exchange optical pumping of 3He at low pressure and high magnetic field," X. Li, **J. Maxwell**, D. Nguyen, *et al.*, Nuclear Instruments and Methods in Physics Research A **1057** (2023). doi:10.1016/j.nima.2023.168792

"Optically pumped polarized 3He++ ion source development for RHIC/EIC," A. Zelenski, *et al.*, Nuclear Instruments and Methods in Physics Research A **1055** (2023). doi:10.1016/j.nima.2023.168494

"Proton spin structure and generalized polarizabilities in the strong quantum chromodynamics regime," D. Ruth, *et al.*, Nature Phys. **18** (2022). doi:10.1038/s41567-022-01781-y

"The PRad Windowless Gas Flow Target," J. Pierce, *et al.*, Nuclear Instruments and Methods in Physics Research A **1003** (2021). doi:10.1016/j.nima.2021.165300

"A concept for polarized He targets for high luminosity scattering experiments in high magnetic field environments," **J. Maxwell**, R. Milner, Nuclear Instruments and Methods in Physics Research A **1012** (2021). doi:10.1016/j.nima.2021.165590

"Enhanced Polarization of Low Pressure ^{3}He through Metastability-Exchange Optical Pumping at High Field," **J. Maxwell**, *et al.*, Nuclear Instruments and Methods in Physics Research A **959** (2020). doi:10.1016/j.nima.2019.02.019

"Proton Form Factor Ratio, $\mu_p G_E^p/G_M^p$ from Double Spin Asymmetry," A. Liyanage, *et al.* (SANE Collaboration), Phys. Rev. C **101**, 035206 (2020). doi:10.1103/PhysRevC.101.035206

"A New Cryogenic Apparatus to Search for the Neutron Electric Dipole Moment," M.W. Ahmed, *et al.*, Journal of Instrumentation, **14** P11017 (2019). doi:10.1088/1748-0221/14/11/P11017

"A small proton charge radius from an electron–proton scattering experiment," W. Xiong, *et al.*, Nature, **575**, 147-150 (2019). doi:10.1038/s41586-019-1721-2

"Revealing Color Forces with Transverse Polarized Electron Scattering," W. Armstrong, *et al.*, Phys. Rev. Letters **122**, 022002 (2019). doi:10.1103/PhysRevLett.122.022002

"Measurements of Non-Singlet Moments of the Nucleon Structure Functions and Comparison to Predictions from Lattice QCD for $Q^2 = 4$ GeV^2," I. Albayrak, *et al.*, Phys. Rev. Lett. **123** 022501 (2019). doi:10.1103/PhysRevLett.123.022501

"Design and Performance of the Spin Asymmetries of the Nucleon Experiment," **J. Maxwell**, *et al.*, Nuclear Instruments and Methods in Physics Research A **885C** (2018). doi:10.1016/j.nima.2017.12.008

"Polarization Transfer Observables in Elastic Electron Proton Scattering at Q^2=2.5, 5.2, 6.8, and 8.5 GeV^2", A.J.R. Puckett, *et al.* Phys. Rev. C **96**, 055203 (2017). doi:10.1103/PhysRevC.96.055203

"Diffusive transfer of polarized ^{3}He gas through depolarizing magnetic gradients," **J. Maxwell**, C. Epstein, R. Milner, Nuclear Instruments and Methods in Physics Research A **777** (2015). doi:10.1016/j.nima.2015.01.004

"Polarization Transfer in Wide-Angle Compton Scattering and Single-Pion Photoproduction from the Proton" C. Fanelli, *et al.* Phys. Rev. Lett. **115** 152001 (2015). doi:10.1103/PhysRevLett.115.152001

"Liquid crystal polarimetry for metastability exchange optical pumping of ^{3}He," **J. Maxwell**, C. Epstein, R. Milner, Nuclear Instruments and Methods in Physics Research A **764** (2014). doi:10.1016/j.nima.2014.07.049

"Dynamically Polarized Target for the g2p and GEp Experiments at Jefferson Lab," J. Pierce, **J. Maxwell**, *et al.*, Nuclear Instruments and Methods in Physics Research A **738** (2013). doi:10.1016/j.nima.2013.12.016

"Search for effects beyond the Born approximation in polarization transfer observables in $\vec{e}p$ elastic scattering" M. Meziane, E. J. Brash, M. K. Jones *et al.* Phys. Rev. Lett. **106** 132501 (2011). doi:10.1103/PhysRevLett.106.132501

"Polarization components in π^0 photoproduction at photon energies up to 5.6 GeV" W. Luo, E. J. Brash, R. Gilman, M. K. Jones *et al.* Phys. Rev. Lett. **108** (2011). doi:10.1103/PhysRevLett.108.222004

"Recoil polarization measurements of the proton electromagnetic form factor ratio to Q 2=8.5 GeV 2" A. J. R. Puckett, E. J. Brash, M. K. Jones, *et al.* Phys. Rev. Lett. **104** 242301 (2010). doi:10.1103/PhysRevLett.104.242301

INVITED TALKS

b1/Azz Tensor Collaboration Meeting, Newport News, VA, October 2025. *"JLab Target Group R&D Progress."*

Polarized Ion Sources and Beams at EIC, Community Meeting, Stony Brook University, Stony Brook, NY, March 2025. *"Search for Exotic Glue in Nuclei: Gluonic Transversity in Polarized DIS."*

Nuclear Physics Seminar, University of Tennessee, Knoxville, TN, November 2024. *"Polarizing Nuclei for Nuclear Scattering Targets."*

International Workshop on Polarized Sources, Targets & Polarimetry, Newport News, VA, September 2024. *"Polarized Gas Targets for Nuclear and Particle Physics."*

Experimental Nuclear Magnetic Resonance Conference, Pacific Grove, CA, April 2024. *"Continuous-wave NMR for Nuclear Physics Polarized Scattering Experiments."*

6th Joint Meeting of the APS Division of Nuclear Physics and the Physical Society of Japan, Hawaii, November 2023. *"A High-Field Polarized 3He Target for Jefferson Lab's CLAS12 Spectrometer."*

25th International Symposium on Spin Physics, Durham, North Carolina, September 2023. *"A High-Field Polarized 3He Target for Jefferson Lab's CLAS12 Spectrometer."*

Workshop on Tensor Spin Observables, Trento, Italy, July 2023. *"A New NMR System for JLab Solid Polarized Targets"* and *"JLab's Polarized NH3 and ND3 Targets."*

New England Section APS Meeting, Durham, New Hampshire, October 2022. *"Polarized Targets at Jefferson Lab: Probing Nuclear Spin Structure."*

Workshop on Exploring QCD with Light Nucei at EIC, Stony Brook, NY, January 2020. *"Polarized ^{3}He for Neutron Structure Measurements."*

International Workshop on Polarized Sources, Targets & Polarimetry, Knoxville, TN, September 2019. *"NMR Measurements for Solid Polarized Targets."*

Physics Department Seminar, University of Virgnia, Charlottesville, VA, February 2019. *"Probing Polarized Nuclei to Explore the Structure of Matter."*

23rd International Symposium on Spin Physics, Ferrara, Italy, October 2018. *"NMR Measurements for Solid Polarized Targets."*

International Workshop on Polarized Sources, Targets & Polarimetry, Daejeon, South Korea, October 2017. *"NMR Polarization Measurements for Jefferson Lab's Solid Polarized Targets."*

Physics Department Colloquium, The Catholic University, Washington, DC, February 2017. *"Phantom Physics: Applications of a Polarized He3 Ion Source for Ghostbusting."*

Electron-Ion Collider User Group Meeting, Argonne National Laboratory, Lemont, IL, July 2016. *"Search for Exotic Gluonic States in the Nucleus."*

Lunch Seminar, Thomas Jefferson National Accelerator Facility, Newport News, VA, June 2016. *"Phantom Physics: Applications of a Polarized He3 Ion Source for Ghostbusting."*

Physic Department Colloquium, College of William and Mary, Williamsburg, VA, November 2015. *"Development of a Polarized Helium-3 Ion Source for an Electron Ion Collider."*

LNS Lunch Seminar, Massachusetts Institute of Technology, Cambridge, MA, May 2015. *"Search for a Neutron Electric Dipole Moment."*

21st International Symposium on Spin Physics, Beijing, China, October 2014. *"A Polarized He3 Ion Source for RHIC and eRHIC."*

International Workshop on Accelerator Science and Technology for Electron-Ion Collider, Newport News, VA, March 2014. *"Development of a Polarized He3 Beam Source for RHIC with EBIS."*

NIST Neutron Physics Group Seminar, Gaithersburg, MD, October 2013. *"Development of a Polarized He3 Beam Source for RHIC."*

International Workshop on Polarized Sources, Targets & Polarimetry, Charlottesville, VA, September 2013. *"Development of a Polarized He3 Beam Source for RHIC."*

LNS Lunch Seminar, Massachusetts Institute of Technology, Cambridge, MA, September 2013. *"Development of a Polarized He3 Beam Source for RHIC using EBIS."*

LNS Special Seminar, Massachusetts Institute of Technology, Cambridge, MA, December 2011. *"Solid Polarized Targets for Nucleon Spin Structure Measurements."*

Nuclear Seminar, University of New Hampshire, Durham, NH, February 2011. *"Solid Polarized Targets for Spin Structure Measurements."*

Polarized Drell-Yan Physics Workshop, Santa Fe, NM, November 2010. *"Radiation Damage in Solid Polarized Targets."*

Hall-C User Group meeting, Jefferson Lab, Newport News, VA, January 2010. *"Polarized Target Performance During SANE."*

PROCEEDINGS

"A High-Field Polarized 3He Target for Jefferson Lab's CLAS12 Spectrometer."
J. Maxwell, Proceedings, 25th International Symposium on Spin Physics (SPIN 2023). doi:10.22323/1.456.0213

"First Use of a Longitudinally Polarized Target with CLAS12."
C. Keith, Proceedings, 19th International Workshop on Polarized Sources, Targets, and Polarimetry (PSTP 2022). doi:10.22323/1.433.0009

"A High-Magnetic-Field Polarized 3He Target for JLab's CLAS12."
J. Maxwell, Proceedings, 19th International Workshop on Polarized Sources, Targets, and Polarimetry (PSTP 2022). doi:10.22323/1.433.0010

"A High-Magnetic-Field Polarized 3He Target for JLab's CLAS12 Spectrometer."
J. Maxwell, JPS Conf.Proc. 37 (2022) 021205. doi:10.7566/JPSCP.37.021205

"Development of a Polarized 3He++ Ion Source for the EIC."
M. Musgrave, Proceedings, 18th International Workshop on Polarized Sources, Targets, and Polarimetry (PSTP 2019). doi:10.22323/1.379.0043

"Developing New Q-meters for NMR Measurements of Polarized Solids."
J. Maxwell, Proceedings, 18th International Workshop on Polarized Sources, Targets, and Polarimetry (PSTP 2019). doi:10.22323/1.379.0051

"The neutron electric dipole moment experiment at the Spallation Neutron Source."
K. Leung, Proceedings International Workshop on Particle physics at Neutron Sources, (2018). arXiv:1903.02700

"Search for Exotic Glue in Nuclei: Gluonic Transversity in Polarized DIS."
J. Maxwell, Proceedings of the 23rd International Spin Physics Symposium, (2018). doi:10.22323/1.346.017

"NMR Measurements for Solid Polarized Targets at Jefferson Lab."
J. Maxwell, Proceedings of the 23rd International Spin Physics Symposium, (2018). doi:10.22323/1.346.0102

"Optically-Pumped Polarized H And 3He Ion Sources Development at RHIC." A. Zelenski *et al.*, Proceedings of the 9th International Particle Accelerator Conference, (2018). doi:10.18429/JACoW-IPAC2018-TUYGBE4

"NMR Polarization Measurements for Jefferson Lab's Solid Polarized Targets." **J. Maxwell**, Proceedings of the 17th International Workshop on Polarized Sources, Targets and Polarimetry, (2017). doi:10.22323/1.324.0011

"Polarized 3He++ Ion Source for RHIC and an EIC." M. Musgrave *et al.*, Proceedings of the 17th International Workshop on Polarized Sources, Targets and Polarimetry, (2017). doi:10.22323/1.324.0020

"Development of a Polarized He3 Beam Source for RHIC using EBIS." **J. Maxwell** *et al.*, International Journal of Modern Physics: Conference Series, Vol. 40, 1660102, (2016). doi:10.1142/S2010194516601022

"Polarized He3 Ion Source for RHIC and eRHIC." **J. Maxwell**, Proceedings of the 16th International Workshop on Polarized Sources, Targets and Polarimetry, (2015). doi:10.22323/1.243.0003

"Development of a Polarized He3 Beam Source for RHIC using EBIS." **J. Maxwell** *et al.*, Physics of Particles and Nuclei, Vol. 45, No. 1, (2014). doi:10.1134/S1063779614010651

"Dynamically polarized sample at the Spallation Neutron Source." J. Pierce *et al.*, Journal of Physics: Conference Series 251 (2010) 012088 (2009). doi:10.1088/1742-6596/251/1/012088

"Radiation Damage and Recovery in Polarized $^{14}NH_3$ Ammonia Targets at Jefferson Lab." **J. Maxwell**, Proceedings of the 13th International Workshop on Polarized Sources, Targets and Polarimetry (2009). doi:10.1142/9789814324922_0018